

Fall 2026 DSC 242: Homework 1

Due 11:59PM Tuesday, October 6

1. (Hoeffding's lemma, 15 points) Suppose the random variable X satisfies that $a \leq X \leq b$ almost surely for two finite constants $a < b$.

(a) Show that $X - \mathbb{E}[X]$ is $\frac{(b-a)^2}{4}$ -subgaussian.

(Hint: Use the fact that $\psi(\lambda) = \int_0^\lambda \int_0^t \psi''(s) ds dt$, and show that $\psi''(\lambda) = \text{Var}[\tilde{X}]$ for a random variable $\tilde{X}$ satisfying also that $a \leq \tilde{X} \leq b$.)

(b) Show that the constant $\frac{(b-a)^2}{4}$ cannot be improved in general: find a random variable X with $a \leq X \leq b$ almost surely such that $X - \mathbb{E}[X]$ is not σ^2 -subgaussian for any $\sigma^2 < \frac{(b-a)^2}{4}$.

(Hint: First show that if a mean-zero random variable Y is σ^2 -subgaussian, then $\text{Var}[Y] \leq \sigma^2$.)

2. (Bernstein's lemma, 15 points) Let X be a random variable satisfying that $\mathbb{E}[X] = 0$, $\text{Var}[X] \leq \sigma^2$, and $|X| \leq b$ almost surely for a finite constant b . Show that for any $\lambda \in (-3/b, 3/b)$,

$$\psi(\lambda) \leq \frac{\lambda^2 \sigma^2}{2(1 - |\lambda|b/3)}.$$

Then further conclude that X is $(2\sigma^2, 2b/3)$ -subexponential.

(Hint: Use the Taylor expansion of $e^{\lambda x}$.)

3. (van Handel 3.3, 20 points) Let X be a mean-zero random variable, and let $X_+ = \max(X, 0)$.

(a) Show that for any $t > 0$,

$$\mathbb{P}[X \geq t] \leq \inf_{p>0} \frac{\mathbb{E}[X_+^p]}{t^p} \leq \inf_{\lambda \geq 0} e^{-\lambda t} \mathbb{E}[e^{\lambda X}],$$

where the first infimum is taken over all real numbers $p > 0$. (Thus a tight bound on all moments of X_+ is at least as good as the Chernoff bound.)

(Hint: For the second inequality, first show that $\mathbb{E}[X_+^p]/t^p \rightarrow \mathbb{P}[X > 0]$ as $p \rightarrow 0_+$.)

(b) The improvement can be dramatic. Let $X = \epsilon Y$, where ϵ is a Rademacher random variable independent of Y , and $\mathbb{P}[Y \geq y] = y^{-\alpha}$ for all $y \geq 1$ with some constant $\alpha > 1$. For $t \geq e$, compute the exact tail probability $\mathbb{P}[X \geq t]$ and both infima in part (a). By what factor does each of the two bounds overestimate $\mathbb{P}[X \geq t]$?

4. (Vershynin 2.5.7, 15 points) Recall the subgaussian norm of a random variable X ,

$$\|X\|_{\psi_2} = \inf \left\{ K > 0 : \mathbb{E}[e^{X^2/K^2}] \leq 2 \right\}.$$

Show that $\|\cdot\|_{\psi_2}$ is indeed a norm on the space of subgaussian random variables, i.e., it satisfies

- (a) (Positive definiteness) $\|X\|_{\psi_2} = 0$ if and only if $X = 0$ with probability 1.
 - (b) (Absolute homogeneity) $\|cX\|_{\psi_2} = |c| \cdot \|X\|_{\psi_2}$ for any $c \in \mathbb{R}$.
 - (c) (Triangle inequality) $\|X + Y\|_{\psi_2} \leq \|X\|_{\psi_2} + \|Y\|_{\psi_2}$.
5. (Bennett's inequality, 15 points) Let X be a random variable satisfying $\mathbb{E}[X] = 0$, $\text{Var}[X] = \sigma^2$, and $X \leq b$ with probability 1.

- (a) Show that for all $\lambda \geq 0$,

$$\log \mathbb{E}[e^{\lambda X}] \leq \frac{\sigma^2(e^{\lambda b} - 1 - \lambda b)}{b^2}.$$

(Hint: You may use the fact that the function $u \mapsto (e^u - 1 - u)/u^2$ is nondecreasing.)

- (b) Further let $X_1, \dots, X_n$ be i.i.d. random variables equal in law to X . Show that for all $t \geq 0$,

$$\mathbb{P}\left[\sum_{i=1}^n X_i \geq t\right] \leq \exp\left(-\frac{n\sigma^2}{b^2} h\left(\frac{bt}{n\sigma^2}\right)\right)$$

where $h(x) = (1+x)\log(1+x) - x$.

6. (How much independence is needed? 20 points) Recall Hoeffding's inequality from Lecture 1: if $X_1, \dots, X_n$ are independent and mean-zero, and each X_i is σ_i^2 -subgaussian, then $\sum_{i=1}^n X_i$ is $(\sum_{i=1}^n \sigma_i^2)$ -subgaussian. This problem examines what remains true when the independence assumption is weakened.

- (a) (No independence) Show that if $X_1, \dots, X_n$ are mean-zero and each X_i is σ_i^2 -subgaussian, then without any assumption on their dependence, $\sum_{i=1}^n X_i$ is $(\sum_{i=1}^n \sigma_i)^2$ -subgaussian. Show also that this cannot be improved in general.
- (b) (Pairwise independence) Let $k \geq 2$ and let $\epsilon_1, \dots, \epsilon_k$ be i.i.d. Rademacher random variables. For every nonempty subset $S \subseteq \{1, \dots, k\}$, define $X_S = \prod_{i \in S} \epsilon_i$. This gives $n = 2^k - 1$ random variables, and we denote their sum by $W = \sum_S X_S$.
 - i. Show that each X_S is a Rademacher random variable, and that X_S and X_T are independent for any $S \neq T$.
 - ii. Find the distribution of W , and show that $\mathbb{P}[W \geq n] = \frac{1}{n+1}$.
 - iii. Conclude that W is not σ^2 -subgaussian for any $\sigma^2 < \frac{n^2}{2 \log(n+1)}$. Compare $\mathbb{P}[W \geq n]$ with the upper bound given by Chebyshev's inequality, and with the bound that Hoeffding's inequality would give if $\{X_S\}$ were independent.